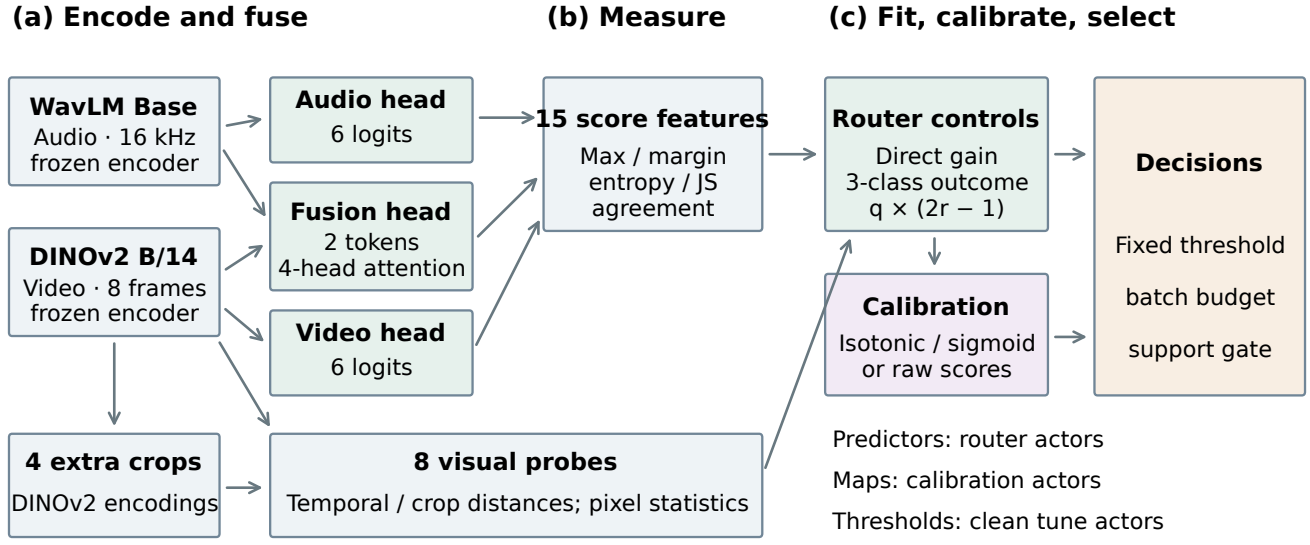
# S1. Implementation: encoders, heads, and features

The expert order is audio, interaction fusion, video. Each six-class logit vector is centered and divided by its own standard deviation, floored at$10^{-8}$, then converted to softmax probabilities. Features are three maxima, three top-two margins, three normalized entropies, three pairwise Jensen–Shannon divergences, and three top-one agreements. Pair order is audio/fusion, audio/video, fusion/video.

The eight added visual measurements are the mean and maximum cosine distance between adjacent frame embeddings; mean and maximum cosine distance between an original frame and its mild crop; central luminance standard deviation; central mean absolute spatial gradient; central mean absolute temporal difference; and central mean luminance. All use decoded 224-square letterboxed RGB frames in[0,1]. Eight frames are uniformly spaced. Four alternate frames are center-cropped at pixels 11:213 and bilinearly resized to 224 square. The central image statistics use pixels 56:168 in each spatial dimension, and luminance is the simple RGB mean.

Frozen WavLM Base yields a 768-dimensional temporal mean; DINOv2 ViT-B/14 yields a 768-dimensional average over frames. Existing expert heads use two 768-to-96 projections, layer normalization and GELU. Interaction fusion uses two modality tokens and four attention heads; the classifier maps the concatenated 192 dimensions through 96 hidden units to 6 classes. The same preprocessing and expert-head checkpoints are retained under visual transformations. Seeds 11 and 22 are both evaluated; they share pretrained encoders and data roles.

All new histogram boosting models have 100 iterations, 7 leaf nodes, minimum leaf size 20, learning rate .05, L2=1, no early stopping and seed 56. Actor-uniform weights are constructed on the router role before disagreement restriction and renormalized for each fitting subset. Direct regression estimates signed gain. The three-class classifier estimates$\Delta \in \{-1, 0, 1\}$. Conditional heads estimate non-neutrality and rescue conditional on non-neutrality. No model uses actor identity, corruption labels, target labels, or clean-reference features at inference.




## Scope of the added head experiment

The added comparison reuses previously trained head weights, and newly executes their predictions on external features. It does not train new foundation encoders. The concatenation head, attention head, and residual control all receive the same frozen modality features. Each of five matching audio and video heads participates in the corresponding checkpoint comparison. Twenty-five checkpoint hashes (five expert kinds times five seeds) identify the exact weights. Model-family counts do not multiply the number of independent actors.

# S2. Dataset selection and source independence

The study has two different evidential roles. CREMA-D's ten original test actors have already been inspected. All new CREMA-D results are exploratory. RAVDESS is an external reserve whose selection, model family, thresholds and evaluation script were locally hashed before feature inference. No RAVDESS examples fit experts, calibration maps, support limits or tuning thresholds.

The CREMA router/calibration/tune roles contain nine actors each. Twenty-four eligible clips per actor are chosen by SHA-256 of a fixed prefix plus clip ID. For these 648 clips, clean, central 25% occlusion, and blur 2 inputs produce 1,944 condition-specific observations. Conditions from the same clip and actor remain dependent; they are not independent calibration sources.

The RAVDESS subset uses all 24 actors and six shared intended-emotion classes. Four full-AV speech clips per actor/class are chosen by a separate fixed filename hash, giving 576 clips. Emotion codes 01, 03, 04, 05, 06, 07 map to neutral, happy, sad, angry, fearful and disgust. Calm and surprise are excluded by the six-class head vocabulary before evaluation. Neither outcome quality nor media appearance participates in selection.

**Integrity and access.** The official Zenodo record and source archive checksums are saved. Downloaded ZIP members are checked with their central-directory CRC32; local SHA-256 values identify the exact media used. Whole-archive MD5 values are provenance metadata, not claimed as verified. The transfer optimization is recorded in a dated pre-inference deviation. The first feature-inference log points to the fitted artifact lock. The locks are local audit evidence, not third-party registration.

**Independence limits.** Actor grouping prevents actor reuse between local roles. It does not prove absence of unknown pretrained-encoder exposure or semantic duplication across Internet corpora. Both datasets measure intended acted expressions. Neither provides physical humanoid execution, natural source occlusion, or a newly collected within-CREMA confirmation set.

## Evaluation targets and transformations

The labels describe performed intended expressions. They are not measurements of private mental state or a robot's conversational understanding. The static gray intervention modifies the center of each decoded, letterboxed image; the moving intervention changes its position across sampled frames. Audio stays fixed. Neither transformation is natural scene occlusion, and the main evidence does not support hidden-object reconstruction. All conditions of an actor remain grouped in source-level analyses.

# S3. Evidence chronology and protocol deviations

## Inspected diagnosis versus external confirmation

The original CREMA test was already inspected when the mechanisms in this paper were analyzed. Its scores, labels, plateau cases, and transformations are therefore diagnostic evidence. External RAVDESS selection and the primary support-versus-margin decision were fixed and locally hashed before the first feature inference. The primary endpoint is clean input, checkpoint 11, nominal 20% clean tuning budget. A positive paired interval was required for local superiority; the resulting interval is negative.

## Source retrieval deviation

The source archive retrieval was optimized to selected members. Official archive checksums are retained as metadata; member CRC and downloaded member SHA-256 values identify the extracted files. This is not a claim that complete remote archives were downloaded and verified. The dated retrieval deviation preceded feature inference. Filename-hash selection was retained.

## Analyses added after inspection

Calibration-subset enumeration, logit-preprocessing comparisons, and the three-head/five-checkpoint study are exploratory. Their fitting roles remain disjoint from test actors, but choices about which analyses to run were made with knowledge of earlier results. Their timestamps cannot make the external dataset untouched again. These comparisons are reported in full rather than using their best row to replace the primary result.

## Corrections that do not rewrite locked outcomes

The practical support rule uses its original calibration-derived distance cutoff. The separate theorem instead requires regions fixed before calibration; its corrected implementation uses router actors for the cutoff and boundary. Its vacuous outcome is reported. A separate audit restores original per-role actor weights before restricting to conditional outcomes. Reproducing the original masks tests this detail.

## Independence claims that remain unavailable

There is no independent encoder replication, fresh collected confirmation population, natural visibility benchmark, or physical humanoid transfer test. Foundation-model pretraining overlap cannot be excluded from these artifacts. Repeated checkpoints, overlapping calibration subsets, and constructed conditions share observations. No reviewer score or acceptance prediction is inferred from the number of analyses.

## Canonical release

One manuscript source and one matching PDF pair define this release. Site copies are byte-checked against those PDFs. The archive retains experimental dependencies under their historical directory identifiers, but does not expose obsolete manuscript versions as the current paper. Those directory identifiers are provenance paths, not competing scientific versions.

# S4. Decision policies and calibrated predictors

For each method with signed score$s$, the positive batch policy considers only disagreement clips with$s > 0$. It sorts by descending score, breaks ties lexicographically by clip ID, and retains the longest actor-weighted prefix within budget. It does not refill the remaining budget with predicted negative or zero gains. For unequal indivisible weights this greedy prefix is not asserted to solve an arbitrary knapsack optimum.

The independent threshold is the smallest nonnegative score level whose strict exceedance mass on clean tuning actors is at most the nominal budget. Whole ties are excluded together. The threshold is frozen under visual shift. This constrains tuning-set change volume, not future population volume.

Support-aware routing standardizes the eight raw reliability features using clean router data, with standard deviations floored at$10^{-5}$. Squared standardized distance is compared with the 95th percentile of clean calibration distances. Outside support the score is set to$-1$ and audio is retained. This is a heuristic rejection rule, not an exchangeability or uncertainty guarantee.

Isotonic maps fit each conditional probability on the calibration role. Sigmoid alternatives fit logistic regression with C=1 to the clipped probability logit; clipping is$[10^{-6}, 1 - 10^{-6}]$. Direct outcome probabilities are calibrated classwise and renormalized before computing rescue minus harm. All three conditions participate in calibration for these new maps; the support reference and operating thresholds use clean clips only.

The original certificate incorrectly shared a cutoff estimated from calibration covariates. The repaired construction replaces only the certificate cutoff with the 95th percentile of clean router distances (20.5336). The empirical support policy keeps its original 29.981 cutoff, preserving the locked results. The score head and median boundary also use router actors only. Evaluate two fixed, disjoint regions on nine untouched calibration actors. For independent actors and $Y_{gj} \in [-1, 1]$, Hoeffding and a two-region union bound subtract $\sqrt{2\log(2/.05)/9} = 0.9054$. The selected action is the entire union of positive-lower-bound regions; subsequent thresholding is not certified. Neither region qualifies. Conditional on fixed training data, calibration actors must be independent draws from the deployment actor population; external shift is not covered. Mutation tests verify that calibration features and labels cannot change the repaired cutoff or score boundary. This implementation check cannot establish those sampling assumptions.

## Equal-volume diagnostics

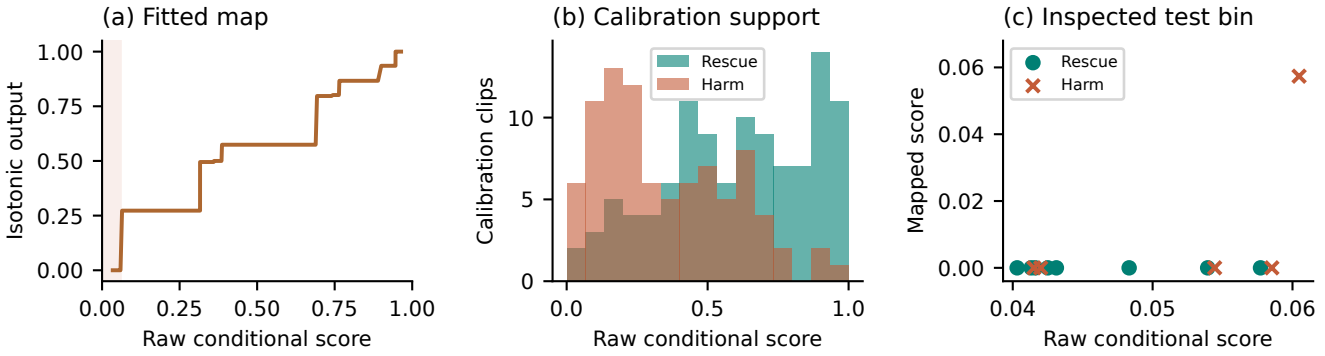
For the new rank-only controls, the positivity filter is deliberately disabled. Scores order disagreements, with clip identifiers breaking ties. The maximal weighted prefix that fits the requested budget is selected. On the balanced 576-clip external set, a nominal 20% budget admits 115 clips (19.965%). This allows comparison of ranking information. It is not the same as a deployable positive-benefit threshold and is labeled separately throughout.

No rank-only interval is claimed for the new head tables. Their fixed-policy intervals resample actor outcomes without retraining, whereas a batch-policy interval would have to re-rank the resampled batch. Leaving that interval blank is preferable to silently applying a fixed-mask bootstrap to a batch algorithm.

# S5. Calibration support and inspected near-zero cases

These 13 cases were diagnosed after the original test was inspected. Eight are rescues. Twelve receive exactly zero conditional rescue probability from an isotonic plateau; none lies outside that map's fitted input range. These labels were not used to refit the new calibration maps. Target and expert predictions are integer class indices in ANG,DIS,FEA,HAP,NEU,SAD order.

| Clip | Gain | Raw r | Mapped r | Rescue p |
|---|---|---|---|---|
| 1001_ITS_DIS_XX | 1 | 0.0431 | 0.0000 | 0.0000 |
| 1001_IWL_DIS_XX | 1 | 0.0483 | 0.0000 | 0.0000 |
| 1020_TAI_NEU_XX | -1 | 0.0605 | 0.0574 | 0.0545 |
| 1036_IWL_FEA_XX | -1 | 0.0585 | 0.0000 | 0.0000 |
| 1036_IWW_ANG_XX | 1 | 0.0539 | 0.0000 | 0.0000 |
| 1058_ITS_NEU_XX | -1 | 0.0415 | 0.0000 | 0.0000 |
| 1058_IWL_NEU_XX | -1 | 0.0544 | 0.0000 | 0.0000 |
| 1058_MTI_SAD_XX | 1 | 0.0425 | 0.0000 | 0.0000 |
| 1058_TAI_SAD_XX | 1 | 0.0413 | 0.0000 | 0.0000 |
| 1058_TIE_FEA_XX | 1 | 0.0416 | 0.0000 | 0.0000 |
| 1060_IEO_ANG_MD | -1 | 0.0420 | 0.0000 | 0.0000 |
| 1060_TIE_SAD_XX | 1 | 0.0403 | 0.0000 | 0.0000 |
| 1085_TIE_ANG_XX | 1 | 0.0577 | 0.0000 | 0.0000 |

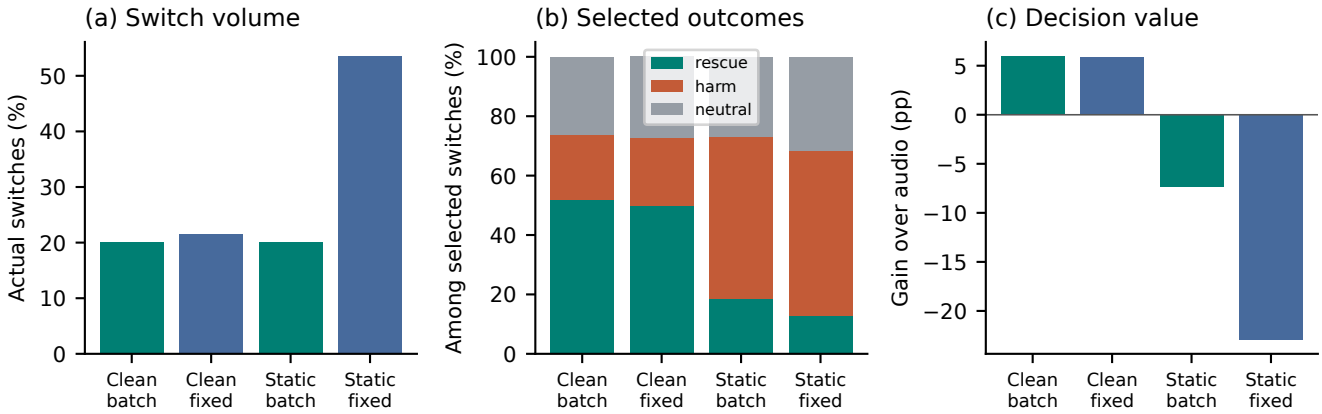



The zero plateau is supported by six non-neutral calibration clips from four actors, all harms. Its empirical conditional outcome is not a population certainty. This support audit distinguishes an internal fitted plateau from out-of-range clipping, and explains why a reliability plot alone is insufficient to locate the failure.

# S6. Switch-volume accounting and paired uncertainty

Let $v$ be change mass and $u$ mean signed gain among changes. The symmetric decomposition$\Delta G = \overline{u}\Delta v + \overline{v}\Delta u$ is exact arithmetic; it is not a causal mediation estimate. The inspected CREMA policy moves from 21.46%to 53.41%changes and from +5.85 to -22.93 points. The volume term is -2.50 points and the selected-quality term is -26.28 points. The corrected positive-budget rule changes none of the selected examples at the original 20%operating point.

| Transition | N | R0 | H0 | R1 | H1 | N1 |
|---|---|---|---|---|---|---|
| retained_switch | 138 | 67 | 34 | 41 | 34 | 63 |
| new_switch | 300 | 21 | 35 | 15 | 210 | 75 |
| abandoned_switch | 38 | 21 | 6 | 7 | 4 | 27 |
| retained_audio | 344 | 24 | 29 | 5 | 97 | 242 |




R0/H0 count rescues/harms before the visual intervention;R1/H1 count them after. N1 is neutral correctness change after intervention. A neutral prediction replacement can have both experts wrong. The mean selector score increases by 0.119, with an exploratory paired actor interval about[0.018,0.223], while correction outcomes worsen.

## Uncertainty unit

For a fixed policy, each actor contributes its mean signed correctness change. Resample actors with replacement and average the resampled means. For a paired comparison use the within-actor difference of two policies, resampling the same actor indices for both. Primary intervals use 5,000 draws; new head-control intervals use 3,000. All are pointwise percentile intervals conditional on trained experts, routing fits, and the selected task.

The identity $G = VQ$ is exact for the reported weighted sample. Its symmetric decomposition describes arithmetic changes between two conditions; it does not identify a causal mediator. Both transformed scores and fusion predictions may change together.

# S7. Full fixed-policy controls: clean external

All rows use 576 clips from 24 actors. Values are percentages or gain points. The nominal clean tuning budget is 20%; actual switching is measured here. Legacy and new predictors have different fitting histories, so their columns are controls, not a pure architecture ablation. Only the clean checkpoint-11 support-versus-margin comparison is primary.

| Seed | Policy | Acc. | Gain | Switch | Rescue | Harm |
|---|---|---|---|---|---|---|
| 11 | margin | 38.54 | 4.17 | 14.06 | 5.56 | 1.39 |
| 11 | confidence | 39.06 | 4.69 | 16.84 | 6.42 | 1.74 |
| 11 | entropy | 38.19 | 3.82 | 19.44 | 6.77 | 2.95 |
| 11 | af_gain | 37.33 | 2.95 | 9.38 | 3.47 | 0.52 |
| 11 | scores_gain | 37.50 | 3.12 | 9.03 | 3.47 | 0.35 |
| 11 | reliable_gain | 34.72 | 0.35 | 2.26 | 0.35 | 0.00 |
| 11 | outcome_raw | 35.24 | 0.87 | 3.82 | 1.04 | 0.17 |
| 11 | outcome_isotonic | 34.38 | 0.00 | 0.35 | 0.00 | 0.00 |
| 11 | two_raw | 36.11 | 1.74 | 5.73 | 1.91 | 0.17 |
| 11 | two_isotonic | 35.24 | 0.87 | 3.82 | 1.04 | 0.17 |
| 11 | two_platt | 35.24 | 0.87 | 3.12 | 0.87 | 0.00 |
| 11 | supported_gain | 34.38 | 0.00 | 0.00 | 0.00 | 0.00 |
| 11 | certified_gain | 34.38 | 0.00 | 0.00 | 0.00 | 0.00 |
| 11 | legacy_opportunity_advantage | 39.24 | 4.86 | 20.31 | 7.81 | 2.95 |
| 11 | legacy_logistic | 38.37 | 3.99 | 18.40 | 6.60 | 2.60 |
| 11 | legacy_gb_gain | 38.89 | 4.51 | 15.97 | 6.42 | 1.91 |
| 11 | legacy_rf_gain | 37.33 | 2.95 | 11.11 | 4.34 | 1.39 |
| 11 | legacy_mlp | 38.37 | 3.99 | 17.71 | 6.94 | 2.95 |
| 22 | margin | 41.49 | 5.90 | 16.67 | 8.33 | 2.43 |
| 22 | confidence | 41.49 | 5.90 | 17.53 | 8.33 | 2.43 |
| 22 | entropy | 41.84 | 6.25 | 17.71 | 8.85 | 2.60 |
| 22 | af_gain | 40.80 | 5.21 | 9.38 | 5.90 | 0.69 |
| 22 | scores_gain | 41.49 | 5.90 | 11.98 | 7.12 | 1.22 |
| 22 | reliable_gain | 41.15 | 5.56 | 10.24 | 6.25 | 0.69 |
| 22 | outcome_raw | 41.49 | 5.90 | 11.81 | 7.12 | 1.22 |
| 22 | outcome_isotonic | 39.41 | 3.82 | 5.56 | 3.99 | 0.17 |
| 22 | two_raw | 41.32 | 5.73 | 15.45 | 7.81 | 2.08 |
| 22 | two_isotonic | 39.93 | 4.34 | 11.28 | 5.73 | 1.39 |
| 22 | two_platt | 41.15 | 5.56 | 14.93 | 7.64 | 2.08 |
| 22 | supported_gain | 35.59 | 0.00 | 0.00 | 0.00 | 0.00 |
| 22 | certified_gain | 35.59 | 0.00 | 0.00 | 0.00 | 0.00 |

**Endpoint interpretation.** Support and certificate rows retain audio whenever they reject. Zero switch harm can therefore coincide with zero retained multimodal benefit. Paired differences and actor-level outcomes, rather than separate unpaired accuracy intervals, determine the primary comparison.

# S8. Full fixed-policy controls: static occluder

All rows use 576 clips from 24 actors. Values are percentages or gain points. The nominal clean tuning budget is 20%; actual switching is measured here. Legacy and new predictors have different fitting histories, so their columns are controls, not a pure architecture ablation. Only the clean checkpoint-11 support-versus-margin comparison is primary.

| Seed | Policy | Acc. | Gain | Switch | Rescue | Harm |
|---|---|---|---|---|---|---|
| 11 | margin | 30.38 | -3.99 | 34.55 | 6.08 | 10.07 |
| 11 | confidence | 29.69 | -4.69 | 36.81 | 6.42 | 11.11 |
| 11 | entropy | 27.43 | -6.94 | 41.84 | 6.60 | 13.54 |
| 11 | af_gain | 32.47 | -1.91 | 20.83 | 3.99 | 5.90 |
| 11 | scores_gain | 33.16 | -1.22 | 14.41 | 2.60 | 3.82 |
| 11 | reliable_gain | 32.64 | -1.74 | 8.68 | 0.87 | 2.60 |
| 11 | outcome_raw | 32.12 | -2.26 | 9.55 | 0.69 | 2.95 |
| 11 | outcome_isotonic | 34.55 | 0.17 | 1.04 | 0.17 | 0.00 |
| 11 | two_raw | 32.81 | -1.56 | 11.63 | 1.74 | 3.30 |
| 11 | two_isotonic | 33.33 | -1.04 | 8.51 | 1.22 | 2.26 |
| 11 | two_platt | 33.68 | -0.69 | 6.25 | 0.52 | 1.22 |
| 11 | supported_gain | 34.38 | 0.00 | 0.00 | 0.00 | 0.00 |
| 11 | certified_gain | 34.38 | 0.00 | 0.00 | 0.00 | 0.00 |
| 11 | legacy_opportunity_advantage | 26.74 | -7.64 | 52.95 | 8.51 | 16.15 |
| 11 | legacy_logistic | 27.60 | -6.77 | 46.88 | 7.64 | 14.41 |
| 11 | legacy_gb_gain | 29.34 | -5.03 | 40.28 | 6.77 | 11.81 |
| 11 | legacy_rf_gain | 31.77 | -2.60 | 28.47 | 5.21 | 7.81 |
| 11 | legacy_mlp | 29.17 | -5.21 | 53.30 | 9.38 | 14.58 |
| 22 | margin | 29.51 | -6.08 | 58.16 | 11.81 | 17.88 |
| 22 | confidence | 30.21 | -5.38 | 55.90 | 11.46 | 16.84 |
| 22 | entropy | 29.51 | -6.08 | 55.90 | 11.28 | 17.36 |
| 22 | af_gain | 33.68 | -1.91 | 23.44 | 4.69 | 6.60 |
| 22 | scores_gain | 30.21 | -5.38 | 50.17 | 11.46 | 16.84 |
| 22 | reliable_gain | 34.03 | -1.56 | 23.61 | 6.60 | 8.16 |
| 22 | outcome_raw | 34.38 | -1.22 | 19.79 | 5.38 | 6.60 |
| 22 | outcome_isotonic | 36.63 | 1.04 | 10.76 | 3.65 | 2.60 |
| 22 | two_raw | 35.42 | -0.17 | 17.01 | 5.38 | 5.56 |
| 22 | two_isotonic | 35.42 | -0.17 | 12.50 | 3.82 | 3.99 |
| 22 | two_platt | 35.07 | -0.52 | 16.32 | 5.03 | 5.56 |
| 22 | supported_gain | 35.59 | 0.00 | 0.00 | 0.00 | 0.00 |
| 22 | certified_gain | 35.59 | 0.00 | 0.00 | 0.00 | 0.00 |

**Endpoint interpretation.** Support and certificate rows retain audio whenever they reject. Zero switch harm can therefore coincide with zero retained multimodal benefit. Paired differences and actor-level outcomes, rather than separate unpaired accuracy intervals, determine the primary comparison.

# S9. Full fixed-policy controls: moving occluder

All rows use 576 clips from 24 actors. Values are percentages or gain points. The nominal clean tuning budget is 20%; actual switching is measured here. Legacy and new predictors have different fitting histories, so their columns are controls, not a pure architecture ablation. Only the clean checkpoint-11 support-versus-margin comparison is primary.

| Seed | Policy | Acc. | Gain | Switch | Rescue | Harm |
|---|---|---|---|---|---|---|
| 11 | margin | 32.81 | -1.56 | 22.40 | 4.51 | 6.08 |
| 11 | confidence | 32.47 | -1.91 | 24.48 | 4.51 | 6.42 |
| 11 | entropy | 30.21 | -4.17 | 26.22 | 4.34 | 8.51 |
| 11 | af_gain | 33.16 | -1.22 | 12.15 | 1.74 | 2.95 |
| 11 | scores_gain | 33.16 | -1.22 | 12.85 | 1.91 | 3.12 |
| 11 | reliable_gain | 34.03 | -0.35 | 2.78 | 0.35 | 0.69 |
| 11 | outcome_raw | 32.99 | -1.39 | 9.38 | 1.04 | 2.43 |
| 11 | outcome_isotonic | 34.20 | -0.17 | 0.35 | 0.00 | 0.17 |
| 11 | two_raw | 34.03 | -0.35 | 11.28 | 1.91 | 2.26 |
| 11 | two_isotonic | 33.68 | -0.69 | 7.99 | 1.04 | 1.74 |
| 11 | two_platt | 33.68 | -0.69 | 6.42 | 0.69 | 1.39 |
| 11 | supported_gain | 34.38 | 0.00 | 0.00 | 0.00 | 0.00 |
| 11 | certified_gain | 34.38 | 0.00 | 0.00 | 0.00 | 0.00 |
| 11 | legacy_opportunity_advantage | 28.99 | -5.38 | 45.49 | 7.12 | 12.50 |
| 11 | legacy_logistic | 30.90 | -3.47 | 35.59 | 6.42 | 9.90 |
| 11 | legacy_gb_gain | 31.60 | -2.78 | 29.51 | 4.34 | 7.12 |
| 11 | legacy_rf_gain | 33.68 | -0.69 | 8.68 | 1.04 | 1.74 |
| 11 | legacy_mlp | 30.73 | -3.65 | 42.01 | 7.81 | 11.46 |
| 22 | margin | 33.85 | -1.74 | 17.53 | 4.34 | 6.08 |
| 22 | confidence | 33.51 | -2.08 | 17.88 | 4.34 | 6.42 |
| 22 | entropy | 33.85 | -1.74 | 18.40 | 4.34 | 6.08 |
| 22 | af_gain | 34.03 | -1.56 | 14.93 | 2.78 | 4.34 |
| 22 | scores_gain | 33.33 | -2.26 | 34.55 | 9.03 | 11.28 |
| 22 | reliable_gain | 33.16 | -2.43 | 36.46 | 9.90 | 12.33 |
| 22 | outcome_raw | 32.81 | -2.78 | 30.56 | 7.81 | 10.59 |
| 22 | outcome_isotonic | 34.90 | -0.69 | 18.23 | 4.51 | 5.21 |
| 22 | two_raw | 31.25 | -4.34 | 41.67 | 10.24 | 14.58 |
| 22 | two_isotonic | 31.94 | -3.65 | 35.76 | 8.68 | 12.33 |
| 22 | two_platt | 31.08 | -4.51 | 41.32 | 10.07 | 14.58 |
| 22 | supported_gain | 35.59 | 0.00 | 0.00 | 0.00 | 0.00 |
| 22 | certified_gain | 35.59 | 0.00 | 0.00 | 0.00 | 0.00 |

**Endpoint interpretation.** Support and certificate rows retain audio whenever they reject. Zero switch harm can therefore coincide with zero retained multimodal benefit. Paired differences and actor-level outcomes, rather than separate unpaired accuracy intervals, determine the primary comparison.

# S10. Budget sensitivity at the primary checkpoint

This table varies the nominal clean tuning budget while keeping all fitting roles fixed. Values are gain points or percentages. Every row uses 24 external actors; budgets are not new independent observations. Full CSVs also contain checkpoint 22, all eighteen policies, batch rules, and endpoints.

| Input | Budget | Policy | Gain | Switch | Rescue | Harm |
|---|---|---|---|---|---|---|
| Clean | 5 | Margin | 0.52 | 2.43 | 0.87 | 0.35 |
| Clean | 5 | Direct | 0.17 | 1.91 | 0.17 | 0.00 |
| Clean | 5 | Structured | 0.00 | 0.00 | 0.00 | 0.00 |
| Clean | 10 | Margin | 2.95 | 7.12 | 3.65 | 0.69 |
| Clean | 10 | Direct | 2.60 | 6.42 | 2.60 | 0.00 |
| Clean | 10 | Structured | 0.00 | 0.00 | 0.00 | 0.00 |
| Clean | 20 | Margin | 4.17 | 14.06 | 5.56 | 1.39 |
| Clean | 20 | Direct | 3.12 | 9.03 | 3.47 | 0.35 |
| Clean | 20 | Structured | 0.87 | 3.82 | 1.04 | 0.17 |
| Clean | 30 | Margin | 4.69 | 19.27 | 7.12 | 2.43 |
| Clean | 30 | Direct | 3.12 | 9.03 | 3.47 | 0.35 |
| Clean | 30 | Structured | 0.87 | 3.82 | 1.04 | 0.17 |
| Clean | 40 | Margin | 4.69 | 19.27 | 7.12 | 2.43 |
| Clean | 40 | Direct | 3.12 | 9.03 | 3.47 | 0.35 |
| Clean | 40 | Structured | 0.87 | 3.82 | 1.04 | 0.17 |
| Static | 5 | Margin | 0.00 | 7.81 | 1.22 | 1.22 |
| Static | 5 | Direct | -1.04 | 4.69 | 0.35 | 1.39 |
| Static | 5 | Structured | -0.17 | 0.35 | 0.00 | 0.17 |
| Static | 10 | Margin | -0.69 | 15.97 | 3.12 | 3.82 |
| Static | 10 | Direct | -1.56 | 9.20 | 1.39 | 2.95 |
| Static | 10 | Structured | -0.35 | 0.87 | 0.00 | 0.35 |
| Static | 20 | Margin | -3.99 | 34.55 | 6.08 | 10.07 |
| Static | 20 | Direct | -1.22 | 14.41 | 2.60 | 3.82 |
| Static | 20 | Structured | -1.04 | 8.51 | 1.22 | 2.26 |
| Static | 30 | Margin | -3.99 | 40.62 | 7.64 | 11.63 |
| Static | 30 | Direct | -1.22 | 14.41 | 2.60 | 3.82 |
| Static | 30 | Structured | -1.04 | 8.51 | 1.22 | 2.26 |
| Static | 40 | Margin | -3.99 | 40.62 | 7.64 | 11.63 |
| Static | 40 | Direct | -1.22 | 14.41 | 2.60 | 3.82 |
| Static | 40 | Structured | -1.04 | 8.51 | 1.22 | 2.26 |
| Moving | 5 | Margin | -0.35 | 3.12 | 0.35 | 0.69 |
| Moving | 5 | Direct | -0.35 | 2.26 | 0.17 | 0.52 |
| Moving | 5 | Structured | 0.00 | 0.17 | 0.00 | 0.00 |
| Moving | 10 | Margin | -0.52 | 8.51 | 1.56 | 2.08 |
| Moving | 10 | Direct | -0.35 | 8.16 | 1.56 | 1.91 |
| Moving | 10 | Structured | 0.17 | 0.52 | 0.17 | 0.00 |
| Moving | 20 | Margin | -1.56 | 22.40 | 4.51 | 6.08 |
| Moving | 20 | Direct | -1.22 | 12.85 | 1.91 | 3.12 |
| Moving | 20 | Structured | -0.69 | 7.99 | 1.04 | 1.74 |
| Moving | 30 | Margin | -1.91 | 31.60 | 6.08 | 7.99 |
| Moving | 30 | Direct | -1.22 | 12.85 | 1.91 | 3.12 |
| Moving | 30 | Structured | -0.69 | 7.99 | 1.04 | 1.74 |
| Moving | 40 | Margin | -1.91 | 31.60 | 6.08 | 7.99 |
| Moving | 40 | Direct | -1.22 | 12.85 | 1.91 | 3.12 |
| Moving | 40 | Structured | -0.69 | 7.99 | 1.04 | 1.74 |

The fixed rule selects a threshold on clean tuning actors. On a changed population its actual switch fraction can substantially exceed the nominal rate. The rank-only and positive-batch policies have different definitions and must not be substituted for these rows.

# S11. Actor outcomes and primary paired comparison

Each row is one of the 24 independent-source units under the working independent-actor assumption. Values are gain over audio in points. S is the support-aware rule; M is margin. All actors, including adverse outcomes, remain in every condition.

| Actor | Clean S | Clean M | Static S | Static M | Moving S | Moving M |
|---|---|---|---|---|---|---|
| 01 | 0.00 | 12.50 | 0.00 | -25.00 | 0.00 | -25.00 |
| 02 | 0.00 | 0.00 | 0.00 | 12.50 | 0.00 | 8.33 |
| 03 | 0.00 | 0.00 | 0.00 | -4.17 | 0.00 | 0.00 |
| 04 | 0.00 | 0.00 | 0.00 | -4.17 | 0.00 | -4.17 |
| 05 | 0.00 | 8.33 | 0.00 | -4.17 | 0.00 | -4.17 |
| 06 | 0.00 | 4.17 | 0.00 | -8.33 | 0.00 | 0.00 |
| 07 | 0.00 | 16.67 | 0.00 | 4.17 | 0.00 | 4.17 |
| 08 | 0.00 | 8.33 | 0.00 | -8.33 | 0.00 | -4.17 |
| 09 | 0.00 | 8.33 | 0.00 | -16.67 | 0.00 | -8.33 |
| 10 | 0.00 | 0.00 | 0.00 | -12.50 | 0.00 | 0.00 |
| 11 | 0.00 | 0.00 | 0.00 | 8.33 | 0.00 | 0.00 |
| 12 | 0.00 | -4.17 | 0.00 | -4.17 | 0.00 | 4.17 |
| 13 | 0.00 | 8.33 | 0.00 | 4.17 | 0.00 | 4.17 |
| 14 | 0.00 | -4.17 | 0.00 | -12.50 | 0.00 | -8.33 |
| 15 | 0.00 | -4.17 | 0.00 | -8.33 | 0.00 | -4.17 |
| 16 | 0.00 | 8.33 | 0.00 | 4.17 | 0.00 | 4.17 |
| 17 | 0.00 | 0.00 | 0.00 | -12.50 | 0.00 | -4.17 |
| 18 | 0.00 | 8.33 | 0.00 | -4.17 | 0.00 | 4.17 |
| 19 | 0.00 | 8.33 | 0.00 | 0.00 | 0.00 | 8.33 |
| 20 | 0.00 | 8.33 | 0.00 | 0.00 | 0.00 | -4.17 |
| 21 | 0.00 | 8.33 | 0.00 | 0.00 | 0.00 | 0.00 |
| 22 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | -12.50 |
| 23 | 0.00 | -4.17 | 0.00 | -8.33 | 0.00 | -4.17 |
| 24 | 0.00 | 8.33 | 0.00 | 4.17 | 0.00 | 8.33 |

The primary difference S minus M is $-4.17$ points, with paired 95% interval $[-6.42, -1.91]$. Under static occlusion it becomes $+3.99$ points, interval $[0.69, 7.47]$, because support retains audio while margin can harm it. These two facts describe the clean opportunity cost and corrupted-input protection together. The latter does not replace the failed clean primary test.

Clips within actors are dependent. Resampling individual clips as if independent would change the uncertainty target. All reported paired intervals preserve the source grouping. An all-zero source vector yields a degenerate empirical bootstrap interval, not a guarantee of zero future harm.

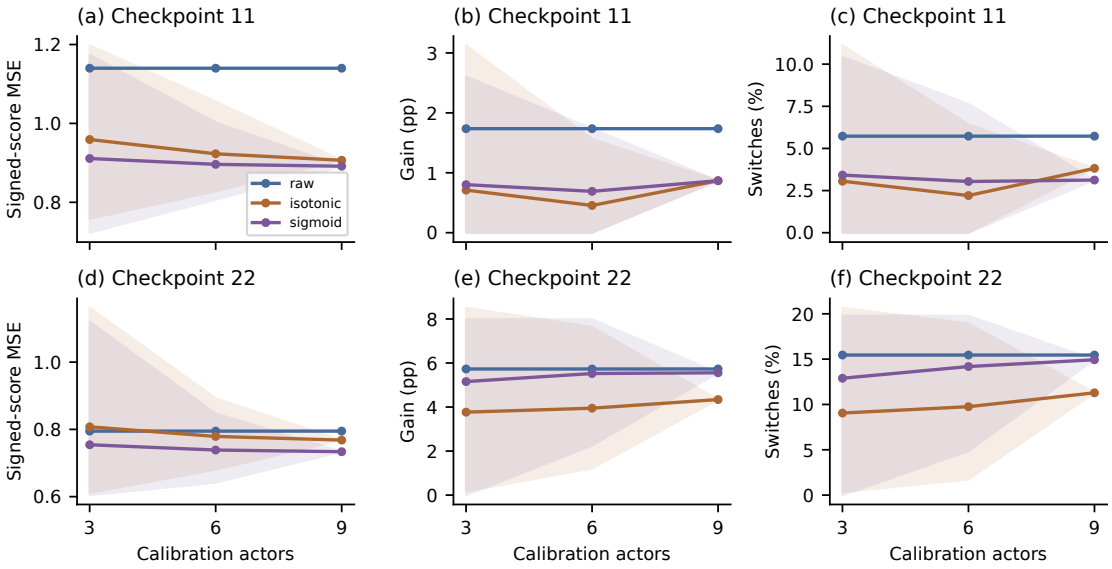
## S12. Calibration-composition sensitivity

All subsets of sizes three and six from the nine original calibration actors are evaluated: 84 each, plus the full nine actors. Trained q/r heads are frozen. Isotonic and sigmoid maps use original role-weighted disagreement data; the conditional rescue map uses only non-neutral disagreements. Single-class calibration subsets use their constant empirical probability, explicitly retained. Raw scores do not depend on the calibration subset and repeat as a reference, not an additional fitted experiment.

The clean tuning role chooses each nonnegative threshold under the same nominal 20% cap. Actual test coverage is measured, not forced. Results below are clean RAVDESS (24 actors), already inspected before this audit. Range means calibration-subset minimum and maximum, not a confidence interval. Gain and coverage are percentage points/percent; MSE is signed-score error on disagreements.

| Seed | Actors | Calibrator | Mean gain | Subset range | Coverage | MSE |
|---|---|---|---|---|---|---|
| 11 | 3 | raw | 1.74 | [1.74,1.74] | 5.73 | 1.140 |
| 11 | 3 | isotonic | 0.71 | [0.00,3.12] | 3.06 | 0.959 |
| 11 | 3 | sigmoid | 0.80 | [0.00,2.60] | 3.42 | 0.911 |
| 11 | 6 | raw | 1.74 | [1.74,1.74] | 5.73 | 1.140 |
| 11 | 6 | isotonic | 0.45 | [0.00,1.56] | 2.21 | 0.923 |
| 11 | 6 | sigmoid | 0.69 | [0.00,1.74] | 3.04 | 0.896 |
| 11 | 9 | raw | 1.74 | [1.74,1.74] | 5.73 | 1.140 |
| 11 | 9 | isotonic | 0.87 | [0.87,0.87] | 3.82 | 0.907 |
| 11 | 9 | sigmoid | 0.87 | [0.87,0.87] | 3.12 | 0.891 |
| 22 | 3 | raw | 5.73 | [5.73,5.73] | 15.45 | 0.795 |
| 22 | 3 | isotonic | 3.77 | [0.17,8.51] | 9.05 | 0.808 |
| 22 | 3 | sigmoid | 5.16 | [0.00,7.99] | 12.89 | 0.754 |
| 22 | 6 | raw | 5.73 | [5.73,5.73] | 15.45 | 0.795 |
| 22 | 6 | isotonic | 3.95 | [1.22,7.64] | 9.75 | 0.779 |
| 22 | 6 | sigmoid | 5.52 | [2.26,7.99] | 14.18 | 0.739 |
| 22 | 9 | raw | 5.73 | [5.73,5.73] | 15.45 | 0.795 |
| 22 | 9 | isotonic | 4.34 | [4.34,4.34] | 11.28 | 0.768 |
| 22 | 9 | sigmoid | 5.56 | [5.56,5.56] | 14.93 | 0.734 |

**Interpretation.** Better score error does not guarantee better frozen-threshold decisions. The checkpoint-11 sigmoid error decreases from 1.140 to 0.891 with all nine actors, but its switch gain decreases from 1.74 to 0.87 points. Checkpoint 22 has a smaller gain decrease. Coverage differs; no matched-rate victory follows. Full data include all five conditions per checkpoint, proper binary Brier scores for rescue/harm, operating-region bias, and zero-rescue fractions. There are 5,070 metric rows but only two fixed head checkpoints and the original evaluation actors.



## S13. Calibration across every external condition

Mean gains and full subset ranges are shown. There are 84 subsets each for sizes three and six; the full nine-actor set has one. Ranges are sensitivity extrema, not confidence intervals. These rows reuse the same 24 external actors.

| Seed | Input | Actors | Map | Gain | Subset range | MSE |
|---|---|---|---|---|---|---|
| 11 | Clean | 3 | isotonic | 0.71 | [0.00, 3.12] | 0.959 |
| 11 | Clean | 3 | sigmoid | 0.80 | [0.00, 2.60] | 0.911 |
| 11 | Clean | 6 | isotonic | 0.45 | [0.00, 1.56] | 0.923 |
| 11 | Clean | 6 | sigmoid | 0.69 | [0.00, 1.74] | 0.896 |
| 11 | Clean | 9 | isotonic | 0.87 | [0.87, 0.87] | 0.907 |
| 11 | Clean | 9 | sigmoid | 0.87 | [0.87, 0.87] | 0.891 |
| 11 | Static | 3 | isotonic | -0.92 | [-3.47, 0.00] | 0.627 |
| 11 | Static | 3 | sigmoid | -1.00 | [-3.47, 0.00] | 0.561 |
| 11 | Static | 6 | isotonic | -0.60 | [-2.43, 0.00] | 0.600 |
| 11 | Static | 6 | sigmoid | -0.82 | [-2.95, 0.00] | 0.552 |
| 11 | Static | 9 | isotonic | -1.04 | [-1.04, -1.04] | 0.588 |
| 11 | Static | 9 | sigmoid | -0.69 | [-0.69, -0.69] | 0.550 |
| 11 | Moving | 3 | isotonic | -0.51 | [-2.26, 0.17] | 0.567 |
| 11 | Moving | 3 | sigmoid | -0.47 | [-1.91, 0.17] | 0.551 |
| 11 | Moving | 6 | isotonic | -0.40 | [-1.04, 0.17] | 0.542 |
| 11 | Moving | 6 | sigmoid | -0.51 | [-1.22, 0.00] | 0.541 |
| 11 | Moving | 9 | isotonic | -0.69 | [-0.69, -0.69] | 0.531 |
| 11 | Moving | 9 | sigmoid | -0.69 | [-0.69, -0.69] | 0.538 |
| 22 | Clean | 3 | isotonic | 3.77 | [0.17, 8.51] | 0.808 |
| 22 | Clean | 3 | sigmoid | 5.16 | [0.00, 7.99] | 0.754 |
| 22 | Clean | 6 | isotonic | 3.95 | [1.22, 7.64] | 0.779 |
| 22 | Clean | 6 | sigmoid | 5.52 | [2.26, 7.99] | 0.739 |
| 22 | Clean | 9 | isotonic | 4.34 | [4.34, 4.34] | 0.768 |
| 22 | Clean | 9 | sigmoid | 5.56 | [5.56, 5.56] | 0.734 |
| 22 | Static | 3 | isotonic | -0.51 | [-2.43, 0.17] | 0.611 |
| 22 | Static | 3 | sigmoid | -0.74 | [-1.91, 0.52] | 0.577 |
| 22 | Static | 6 | isotonic | -0.33 | [-2.43, 0.00] | 0.585 |
| 22 | Static | 6 | sigmoid | -0.68 | [-1.91, 0.35] | 0.571 |
| 22 | Static | 9 | isotonic | -0.17 | [-0.17, -0.17] | 0.575 |
| 22 | Static | 9 | sigmoid | -0.52 | [-0.52, -0.52] | 0.569 |
| 22 | Moving | 3 | isotonic | -3.27 | [-4.69, -0.87] | 0.749 |
| 22 | Moving | 3 | sigmoid | -3.63 | [-4.86, 0.00] | 0.669 |
| 22 | Moving | 6 | isotonic | -3.55 | [-4.69, -2.60] | 0.721 |
| 22 | Moving | 6 | sigmoid | -3.98 | [-4.69, -2.26] | 0.657 |
| 22 | Moving | 9 | isotonic | -3.65 | [-3.65, -3.65] | 0.707 |
| 22 | Moving | 9 | sigmoid | -4.51 | [-4.51, -4.51] | 0.654 |

No subset is selected for deployment on the basis of these outcomes. Lower disagreement-wide score error can accompany lower gain or lower coverage. The released 5,070-row table additionally includes the inspected CREMA conditions, binary rescue/harm Brier scores, operating-region bias, and zero-rescue fractions.

# S14. Fixed-region certificate: proof and scope

The repaired regions are completely fixed on router actors. Conditional on those actors, define $Y_{gj} = N_g^{-1} \sum_{i \in g} \Delta_i 1\{Z_i \in R_j\}$. Dependence among clips within an actor is permitted: the concentration assumption applies to the independent bounded actor means. For each of two regions, $\Pr(\bar{Y}_j - \mathbb{E}Y_j > \epsilon) \leq \exp(-n\epsilon^2/2)$. Substituting $\epsilon = \sqrt{2\log(2/\alpha)/n}$ and applying a union bound proves simultaneous validity. Consequently selecting disjoint positive-bound regions after seeing their bounds is allowed. Choosing the regions themselves from calibration inputs was not justified by that argument.

| Seed | Region | Actor mean | Penalty | Lower bound |
|---|---|---|---|---|
| 11 | 0 | .02778 | .90540 | -.87762 |
| 11 | 1 | .01852 | .90540 | -.88688 |
| 22 | 0 | .02778 | .90540 | -.87762 |
| 22 | 1 | .04630 | .90540 | -.85910 |

All four cells abstain. Correctness of the repaired bound provides no useful certified coverage here. Its population assumptions remain unverified by deterministic actor roles, and it does not certify RAVDESS transfer.

**Mutation check.** Replacing every calibration reliability vector with 999 and changing every calibration label leaves the router-only region parameters identical. Actor-role intersections are empty. We retain original practical-policy cutoffs and its locked result tables; this correction does not turn the failed method into a successful one.

**Coverage and harm intervals.** A new CSV supplies 4,000 actor-bootstrap replicates for each frozen rule's gain, coverage, rescue and harm fractions. Each row states its independent-source count: ten for inspected CREMA, twenty-four for RAVDESS. These are pointwise percentile intervals conditional on fitted models, not simultaneous coverage across all policies and conditions. Degenerate all-zero intervals describe a constant empirical sample, not proof of zero future risk. Earlier batch-policy analyses retain their required re-ranking; these new intervals concern fixed independent thresholds only.

## Why selection among regions is permitted

On the simultaneous event, each selected region has nonnegative population gain because its empirical mean exceeds the common penalty. The regions are disjoint, so the union's gain is their sum. This argument permits choosing which pre-fixed regions to retain after calibration. It does not permit choosing an arbitrary new subset of a region or optimizing its boundary on the same calibration outcomes.

## Why this is not an arbitrary-shift guarantee

The expected actor mean in the bound concerns the same actor-generating population as the independent calibration draws. Changing domains can invalidate that relation. A deterministic implementation check cannot establish population exchangeability. The theorem is standard bounded-mean concentration applied to fixed regions, not the claimed methodological novelty of this study.

# S15. Preprocessing and equal-volume controls

The same fifteen statistics and direct regressor are refit with three logit transforms. Standardization uses within-clip expert mean and standard deviation. Raw softmax preserves logit scale. Temperature softmax fits one positive temperature per expert on original calibration actors, minimizing actor-weighted cross entropy over the three development conditions; log temperature is bounded to $[-4, 4]$. All methods fit routing outcomes on router actors and freeze thresholds using clean tune actors. Temperature fitting consumes calibration labels and is not an information-free transform.

Values below compare fixed-threshold gain and switching with rank-only matched gain at 19.97% actual switching. All rows are exploratory and use the same external clips.

| Seed | Input | Transform | Fixed G | Fixed V | Matched G |
|---|---|---|---|---|---|
| 11 | Clean | standardized | 3.12 | 9.03 | 3.99 |
| 11 | Clean | raw | 2.60 | 7.29 | 5.56 |
| 11 | Clean | temperature | 2.26 | 5.56 | 6.25 |
| 11 | Static | standardized | -1.22 | 14.41 | -2.08 |
| 11 | Static | raw | -2.78 | 11.81 | -3.99 |
| 11 | Static | temperature | -1.04 | 7.64 | -3.47 |
| 11 | Moving | standardized | -1.22 | 12.85 | -2.26 |
| 11 | Moving | raw | -0.69 | 16.32 | -2.26 |
| 11 | Moving | temperature | -1.91 | 11.63 | -2.95 |
| 22 | Clean | standardized | 5.90 | 11.98 | 7.64 |
| 22 | Clean | raw | 9.55 | 19.79 | 9.72 |
| 22 | Clean | temperature | 8.85 | 18.40 | 8.68 |
| 22 | Static | standardized | -5.38 | 50.17 | -1.39 |
| 22 | Static | raw | -4.17 | 61.11 | -1.74 |
| 22 | Static | temperature | -4.34 | 42.36 | -2.60 |
| 22 | Moving | standardized | -2.26 | 34.55 | -0.52 |
| 22 | Moving | raw | -3.12 | 44.27 | -0.69 |
| 22 | Moving | temperature | -3.12 | 35.59 | -0.87 |

Ten standardized refits reproduce the archived scores to within $10^{-10}$. Raw or temperature scaling can improve clean equal-volume ranking in both checkpoints, but this does not establish a universal fixed-threshold advantage. Full released controls include additional score families at matched 5%, 10%, and 20% rates; no positivity filter is applied in that diagnostic.

The fitted temperatures, thresholds, and preprocessing protocol are stored separately. The external test is not used to fit a temperature. Nonetheless this ablation was chosen after inspection, so a best variant cannot be called independently validated.

# S16. Fusion-head control: late

New external inference using existing trained checkpoints. C: clean; S: static; M: moving. Fixed thresholds are tuned at nominal 20%; intervals use 3,000 actor bootstrap draws over 24 sources. Matched ranking selects 19.97%, without positivity filtering or fixed-mask intervals. Values are gain points or switching percentages.

| Seed | Input | Router | Fixed G | 95% CI | Fixed V | Matched G |
|---|---|---|---|---|---|---|
| 11 | C | margin | 7.12 | [4.69, 9.55] | 18.75 | 7.64 |
| 11 | C | direct | 7.29 | [5.21, 9.55] | 16.67 | 7.64 |
| 11 | C | structured | 4.17 | [2.43, 6.08] | 9.55 | 6.60 |
| 11 | S | margin | -1.04 | [-3.30, 1.22] | 25.17 | -0.52 |
| 11 | S | direct | -0.69 | [-2.43, 1.04] | 12.67 | -2.95 |
| 11 | S | structured | -1.04 | [-2.43, 0.35] | 10.76 | -1.56 |
| 11 | M | margin | -0.35 | [-2.61, 1.74] | 21.35 | -0.69 |
| 11 | M | direct | 0.87 | [-0.52, 2.08] | 13.54 | -1.56 |
| 11 | M | structured | 0.69 | [-0.52, 1.91] | 7.29 | -1.91 |
| 22 | C | margin | 5.90 | [2.60, 9.55] | 13.89 | 7.29 |
| 22 | C | direct | 5.21 | [2.43, 7.81] | 17.19 | 6.60 |
| 22 | C | structured | 5.73 | [2.60, 8.85] | 18.75 | 6.94 |
| 22 | S | margin | -2.95 | [-5.56, -0.69] | 33.16 | 0.69 |
| 22 | S | direct | -2.26 | [-5.03, 0.52] | 45.14 | -0.17 |
| 22 | S | structured | -1.91 | [-4.34, 0.52] | 44.27 | 0.35 |
| 22 | M | margin | -1.91 | [-4.17, 0.35] | 17.36 | -2.60 |
| 22 | M | direct | -2.60 | [-5.38, 0.00] | 21.01 | -2.43 |
| 22 | M | structured | -3.12 | [-5.90, -0.17] | 24.13 | -4.51 |
| 33 | C | margin | 4.34 | [1.74, 7.12] | 15.62 | 5.90 |
| 33 | C | direct | 4.69 | [2.26, 7.12] | 14.58 | 4.86 |
| 33 | C | structured | 0.17 | [0.00, 0.52] | 0.17 | 5.90 |
| 33 | S | margin | -6.94 | [-10.07, -4.17] | 38.89 | -3.65 |
| 33 | S | direct | -3.65 | [-6.25, -1.04] | 22.92 | -2.95 |
| 33 | S | structured | 0.00 | [0.00, 0.00] | 0.00 | -3.12 |
| 33 | M | margin | -3.30 | [-6.42, -0.17] | 24.48 | -3.12 |
| 33 | M | direct | -3.12 | [-5.73, -0.52] | 17.71 | -3.12 |
| 33 | M | structured | 0.00 | [0.00, 0.00] | 0.00 | -2.08 |
| 44 | C | margin | 2.60 | [-0.35, 5.73] | 16.67 | 3.82 |
| 44 | C | direct | 2.43 | [-0.00, 5.21] | 15.97 | 3.47 |
| 44 | C | structured | 1.74 | [-0.69, 4.34] | 14.24 | 2.78 |
| 44 | S | margin | -2.43 | [-5.04, 0.17] | 36.11 | -0.87 |
| 44 | S | direct | -1.56 | [-4.34, 0.87] | 23.26 | -0.52 |
| 44 | S | structured | -0.87 | [-3.30, 1.39] | 17.71 | -1.04 |
| 44 | M | margin | -0.87 | [-3.82, 2.08] | 18.75 | -0.87 |
| 44 | M | direct | -2.60 | [-5.38, 0.00] | 23.09 | -1.74 |
| 44 | M | structured | -2.08 | [-4.86, 0.69] | 18.58 | -2.60 |
| 55 | C | margin | 10.42 | [7.63, 13.37] | 18.58 | 11.28 |
| 55 | C | direct | 13.89 | [9.90, 18.06] | 24.31 | 11.81 |
| 55 | C | structured | 5.38 | [3.12, 7.81] | 7.99 | 11.11 |
| 55 | S | margin | -6.08 | [-9.90, -2.60] | 41.67 | -1.74 |
| 55 | S | direct | -3.82 | [-7.12, -0.35] | 32.12 | -2.95 |
| 55 | S | structured | -2.43 | [-4.51, -0.35] | 13.37 | -2.26 |
| 55 | M | margin | -2.08 | [-5.03, 0.52] | 29.17 | -1.74 |
| 55 | M | direct | -2.60 | [-5.21, -0.17] | 23.96 | -2.26 |
| 55 | M | structured | -0.87 | [-2.43, 0.52] | 8.85 | -0.52 |

All rows share WavLM/DINOv2 encoders and the same external population. These controls test head dependence; they do not establish encoder-independent replication.

# S17. Fusion-head control: interaction

New external inference using existing trained checkpoints. C: clean; S: static; M: moving. Fixed thresholds are tuned at nominal 20%; intervals use 3,000 actor bootstrap draws over 24 sources. Matched ranking selects 19.97%, without positivity filtering or fixed-mask intervals. Values are gain points or switching percentages.

| Seed | Input | Router | Fixed G | 95% CI | Fixed V | Matched G |
|---|---|---|---|---|---|---|
| 11 | C | margin | 4.17 | [1.91, 6.42] | 14.06 | 5.03 |
| 11 | C | direct | 3.12 | [1.22, 5.21] | 9.03 | 3.99 |
| 11 | C | structured | 1.56 | [0.69, 2.60] | 3.65 | 3.47 |
| 11 | S | margin | -3.99 | [-7.29, -0.87] | 34.55 | -0.87 |
| 11 | S | direct | -1.22 | [-3.12, 0.69] | 14.41 | -2.08 |
| 11 | S | structured | -1.04 | [-2.26, 0.00] | 8.33 | -2.26 |
| 11 | M | margin | -1.56 | [-4.69, 1.22] | 22.40 | -1.91 |
| 11 | M | direct | -1.22 | [-2.95, 0.35] | 12.85 | -2.26 |
| 11 | M | structured | -0.52 | [-1.39, 0.35] | 4.86 | -1.39 |
| 22 | C | margin | 5.90 | [2.78, 9.20] | 16.67 | 6.77 |
| 22 | C | direct | 5.90 | [3.30, 8.51] | 11.98 | 7.64 |
| 22 | C | structured | 2.95 | [1.22, 5.03] | 6.08 | 7.99 |
| 22 | S | margin | -6.08 | [-8.85, -3.12] | 58.16 | -0.35 |
| 22 | S | direct | -5.38 | [-8.33, -2.60] | 50.17 | -1.39 |
| 22 | S | structured | -3.12 | [-6.25, 0.00] | 31.94 | 1.04 |
| 22 | M | margin | -1.74 | [-4.34, 0.87] | 17.53 | -1.56 |
| 22 | M | direct | -2.26 | [-6.42, 2.08] | 34.55 | -0.52 |
| 22 | M | structured | -3.12 | [-6.60, 0.17] | 17.71 | -3.82 |
| 33 | C | margin | 4.34 | [1.56, 7.47] | 18.58 | 4.69 |
| 33 | C | direct | 2.78 | [-0.17, 5.73] | 14.41 | 4.86 |
| 33 | C | structured | 0.87 | [-0.35, 2.08] | 4.17 | 4.51 |
| 33 | S | margin | -8.33 | [-11.98, -5.21] | 44.62 | -5.03 |
| 33 | S | direct | -3.99 | [-6.25, -1.74] | 18.75 | -3.65 |
| 33 | S | structured | -0.87 | [-1.56, -0.17] | 1.74 | -3.99 |
| 33 | M | margin | -4.51 | [-7.81, -1.56] | 21.01 | -3.82 |
| 33 | M | direct | -2.60 | [-4.86, -0.35] | 13.72 | -4.69 |
| 33 | M | structured | -1.04 | [-3.30, 1.04] | 6.25 | -1.91 |
| 44 | C | margin | 5.90 | [3.12, 8.85] | 17.36 | 6.25 |
| 44 | C | direct | 5.03 | [2.08, 8.33] | 17.01 | 7.29 |
| 44 | C | structured | 5.03 | [2.26, 7.99] | 15.45 | 7.12 |
| 44 | S | margin | -1.22 | [-3.30, 0.87] | 23.44 | -0.52 |
| 44 | S | direct | -1.56 | [-4.17, 1.04] | 22.22 | -0.69 |
| 44 | S | structured | -0.35 | [-2.78, 2.26] | 20.31 | -0.00 |
| 44 | M | margin | -2.60 | [-5.56, 0.35] | 27.43 | -1.56 |
| 44 | M | direct | -3.47 | [-7.12, 0.17] | 32.64 | -1.91 |
| 44 | M | structured | -2.43 | [-5.03, 0.17] | 25.00 | -1.22 |
| 55 | C | margin | 7.47 | [3.99, 11.63] | 26.04 | 6.25 |
| 55 | C | direct | 5.03 | [1.74, 9.03] | 24.31 | 4.34 |
| 55 | C | structured | 4.69 | [1.56, 8.16] | 14.58 | 5.21 |
| 55 | S | margin | -9.03 | [-12.85, -5.21] | 39.24 | -4.17 |
| 55 | S | direct | -4.69 | [-7.64, -1.56] | 29.17 | -3.30 |
| 55 | S | structured | -2.95 | [-5.03, -0.69] | 17.53 | -2.60 |
| 55 | M | margin | -8.85 | [-12.67, -5.21] | 43.58 | -3.65 |
| 55 | M | direct | -5.38 | [-9.03, -1.73] | 31.08 | -3.30 |
| 55 | M | structured | -2.43 | [-5.03, 0.17] | 15.80 | -3.12 |

All rows share WavLM/DINOv2 encoders and the same external population. These controls test head dependence; they do not establish encoder-independent replication.

# S18. Fusion-head control: capacity_control

New external inference using existing trained checkpoints. C: clean; S: static; M: moving. Fixed thresholds are tuned at nominal 20%; intervals use 3,000 actor bootstrap draws over 24 sources. Matched ranking selects 19.97%, without positivity filtering or fixed-mask intervals. Values are gain points or switching percentages.

| Seed | Input | Router | Fixed G | 95% CI | Fixed V | Matched G |
|---|---|---|---|---|---|---|
| 11 | C | margin | 9.38 | [6.77, 11.81] | 20.66 | 9.55 |
| 11 | C | direct | 6.94 | [4.17, 9.72] | 17.53 | 6.94 |
| 11 | C | structured | 2.43 | [0.69, 4.17] | 6.60 | 6.94 |
| 11 | S | margin | -0.17 | [-3.12, 2.78] | 31.42 | 0.35 |
| 11 | S | direct | -2.26 | [-4.69, 0.00] | 19.97 | -2.26 |
| 11 | S | structured | -2.26 | [-3.82, -0.69] | 9.90 | -2.08 |
| 11 | M | margin | 0.69 | [-1.74, 2.95] | 21.88 | 0.69 |
| 11 | M | direct | -0.87 | [-2.43, 0.69] | 11.81 | -1.22 |
| 11 | M | structured | -0.17 | [-0.69, 0.35] | 2.08 | -1.91 |
| 22 | C | margin | 4.69 | [1.91, 7.64] | 14.93 | 6.08 |
| 22 | C | direct | 5.21 | [1.56, 9.38] | 17.01 | 6.08 |
| 22 | C | structured | 5.56 | [2.43, 8.85] | 16.15 | 6.08 |
| 22 | S | margin | -2.08 | [-4.87, 0.35] | 29.69 | -0.52 |
| 22 | S | direct | -4.51 | [-7.29, -1.91] | 39.06 | -2.08 |
| 22 | S | structured | -4.51 | [-7.47, -1.74] | 42.71 | 0.35 |
| 22 | M | margin | -2.78 | [-4.69, -0.87] | 11.11 | -3.65 |
| 22 | M | direct | -3.12 | [-6.08, -0.00] | 19.79 | -3.12 |
| 22 | M | structured | -2.43 | [-5.21, 0.00] | 17.36 | -2.78 |
| 33 | C | margin | 6.42 | [3.82, 9.03] | 17.36 | 6.60 |
| 33 | C | direct | 4.17 | [1.22, 7.29] | 14.76 | 5.73 |
| 33 | C | structured | 3.65 | [1.04, 6.25] | 13.54 | 4.86 |
| 33 | S | margin | -6.42 | [-9.21, -3.65] | 33.33 | -4.34 |
| 33 | S | direct | -2.26 | [-5.03, 0.35] | 21.53 | -2.60 |
| 33 | S | structured | -1.22 | [-3.99, 1.22] | 15.28 | -1.91 |
| 33 | M | margin | -4.51 | [-7.64, -1.39] | 22.57 | -3.65 |
| 33 | M | direct | -2.95 | [-6.08, -0.17] | 19.44 | -2.78 |
| 33 | M | structured | -2.78 | [-5.56, -0.17] | 15.80 | -2.95 |
| 44 | C | margin | 7.12 | [3.99, 10.07] | 20.31 | 6.94 |
| 44 | C | direct | 4.17 | [1.39, 6.94] | 14.24 | 7.12 |
| 44 | C | structured | 0.87 | [-0.52, 2.08] | 3.82 | 5.90 |
| 44 | S | margin | -2.60 | [-5.90, 0.53] | 44.27 | -0.17 |
| 44 | S | direct | -0.69 | [-3.30, 1.91] | 26.56 | 0.17 |
| 44 | S | structured | -0.52 | [-1.74, 0.52] | 3.47 | 1.56 |
| 44 | M | margin | -1.39 | [-4.86, 2.08] | 22.57 | -1.74 |
| 44 | M | direct | 0.69 | [-2.08, 3.30] | 17.36 | 0.69 |
| 44 | M | structured | -0.17 | [-1.22, 0.87] | 3.12 | 0.52 |
| 55 | C | margin | 11.11 | [7.81, 14.76] | 23.44 | 10.59 |
| 55 | C | direct | 10.24 | [6.60, 13.89] | 30.73 | 7.64 |
| 55 | C | structured | 6.42 | [3.99, 8.68] | 11.11 | 7.99 |
| 55 | S | margin | -6.77 | [-10.76, -2.95] | 42.36 | -1.91 |
| 55 | S | direct | -3.12 | [-5.90, -0.17] | 29.69 | -3.30 |
| 55 | S | structured | -1.56 | [-3.65, 0.35] | 14.76 | -1.39 |
| 55 | M | margin | -4.51 | [-7.81, -1.56] | 38.37 | -2.78 |
| 55 | M | direct | -1.74 | [-4.17, 0.69] | 27.26 | -1.91 |
| 55 | M | structured | -0.69 | [-2.08, 0.52] | 7.81 | -1.22 |

All rows share WavLM/DINOv2 encoders and the same external population. These controls test head dependence; they do not establish encoder-independent replication.

# S19. Timing and qualitative example selection

## Measured costs

The following timings are archived wall-clock component measurements from local feature extraction, in seconds. They are not a controlled hardware benchmark or a robot latency claim. The visual timings include eight sampled frames and four additional crop encodings for the reliability probes. Model loading, video decoding, and server overhead are not represented by these component columns.

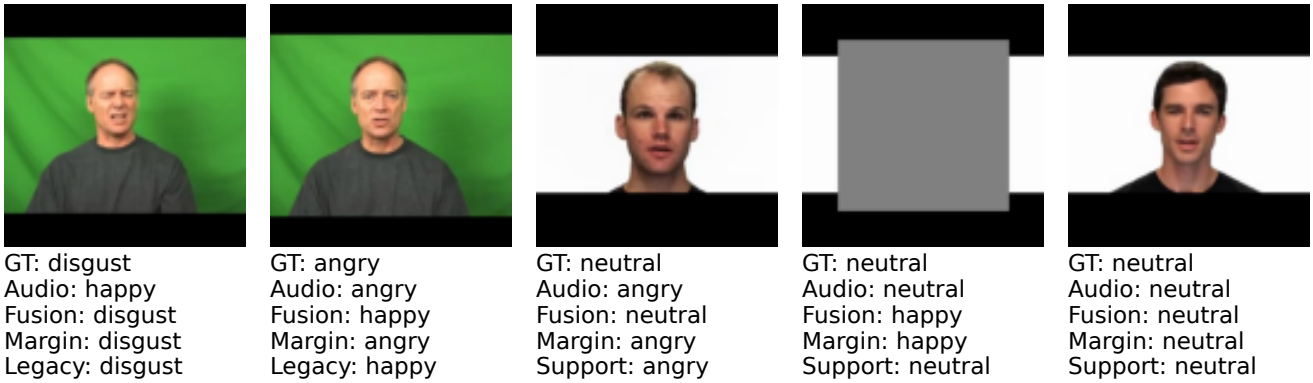
| Component | Observations | Median s | 95th pct s |
|---|---|---|---|
| audio_encoder | 576 | 0.052 | 0.262 |
| visual_encoder_with_reliability_probes | 1728 | 0.383 | 0.559 |

Every router consumes already computed audio and fusion predictions. Consequently this study makes no encoder-compute-saving claim. Cached-logit replay is inexpensive because the encoders have executed previously; its speed cannot be reported as end-to-end video inference speed.

## Selection rules

The teaser is the lexicographically first member of the archived CREMA gallery whose legacy fixed decision retains audio when clean and changes into a harm under the static occluder. The comparison figure chooses the first gallery member in each pre-existing outcome category: beneficial correction, harmful correction, missed rescue, avoided harm, and clean rejection. All compared methods in a column use the same clip and condition. Whole-clip scores determine labels; the illustrated one-second frame does not independently determine them.

The gallery is an inspected subset, so this rule makes selection auditable rather than representative of all clips. It is not a random qualitative sample and cannot estimate frequency. The full numerical tables are the prevalence evidence. Clean and transformed frame scale, ground-truth intended labels, and expert outputs are displayed explicitly.




## Website semantics

Playback replays recorded videos and measured clip-level outputs. It does not stream fresh model updates. Natural scene occlusion and object-memory demonstrations are a separate pilot and do not support this audio–visual intended-emotion study. Source-video attribution and license information remain alongside the demonstrations.

# S20. Reproduction guide and remaining evidence gaps

## Three levels of reproduction

**Artifact checks.** Run the delivered validation script to recompute rescue minus harm identities, verify new prediction arrays against outcomes, compare shared predictions against the earlier checkpoint, and inspect the release hashes. This needs CPU Python, NumPy and pandas. The figure and table scripts regenerate the displayed summaries.

**Frozen-feature head inference.** The complete ZIP supplies selected CREMA and RAVDESS feature caches, the twenty-five small trained head checkpoints and normalization, model definitions, and routing code. Run `outputs/revision59/architecture_` from the archive root. It executes head inference and router fitting, generating 45 prediction archives and 270 policy rows. This is newly computed inference using existing trained heads, not encoder retraining.

**Raw-input extraction.** Official media and pinned WavLM/DINOv2 weights are needed to recompute features. They are not silently downloaded by the portable replay. Source manifests, hashes, extraction scripts and original environment requirements document that route. Preserve source attribution and dataset license conditions. Hardware-specific paths may need adjustment.

## Preprocessing and calibration audits

Run `outputs/revision59/preprocessing.py` for standardized/raw/temperature comparisons and matched-rate controls. Run `outputs/revision58/audit58.py` for exhaustive calibration subsets and the repaired region certificate. The historical directory names are dependency identifiers. Current paper files reside only in `manuscript/`; the site PDF copies must match them byte-for-byte.

Trusted fitted-model files may use Python serialization; only load the supplied local artifacts or other trusted sources. The complete ZIP includes requirement records and checksums. No private API credentials or complete raw dataset is required for the portable analyses.

## What the added evidence changes

Direct fixed-threshold routing loses accuracy under the static intervention for all fifteen head/checkpoint combinations. Margin loses in all fifteen, and structured routing in fourteen with one zero. At the common 19.97% switching rate, direct gain is negative in fourteen of fifteen static cases (mean $-2.05$ points). However, direct mean scores rise in only six of fifteen. The repeated failure is harmful switching under changed evidence; universal confidence inversion is not supported. Preprocessing controls identify discarded confidence information as a sensitivity, not a confirmed remedy.

## What remains unresolved

The study still lacks independent encoder families, a natural-visibility population and a new untouched reserve for a redesigned remedy. The head and preprocessing extensions are exploratory. The support detector fails the locked clean external comparison. The corrected finite-region bound provides no certified switches at nine calibration sources. No result establishes robot transfer, exact hidden-object localization, or a top-tier acceptance score.

The evidence-gap record states these boundaries explicitly. More rows on the same population cannot substitute for independent evidence. A further method paper would need a new mechanism with a validated advantage under an evaluation not used to choose it.